

Adaptive Multimodal Human-Robot Communication Using a Multi-Agent AI Framework with Safety-Aware Decision Making

Dhivya S K

24BRS1111

Sharmista Neravati

24BRS1145

Jithesh Chowdary Palukuri

24BRS1073

The Challenge: Why Current HRI Systems Fall Short

Single-Modal Limitations

Most robots rely on speech-only or task-specific architectures, making them brittle in real-world conditions where noise and ambiguity are unavoidable.

Perception Failures

Noisy environments, speech ambiguity, and occluded objects cause frequent command misinterpretation and execution failures in deployed systems.

Safety Blindspots

Existing systems execute commands without verifying environmental safety—creating real risks of collisions and accidents in human-robot proximity.

The Opportunity: 2.5 Billion People Need Assistive Robots

2.5B

People Today

WHO estimates currently requiring assistive products worldwide

3.5B

By 2050

Projected global demand for assistive solutions within three decades

Collaborative robots are already deployed across **healthcare, manufacturing, logistics, and service sectors**. Natural, intuitive, and safe interaction is no longer optional—it is essential for real-world adoption at scale.

Our Solution: Multi-Agent Multimodal Framework



Voice Agent

Speech recognition via **Whisper** for natural language commands



Gesture Agent

Hand gesture recognition for complementary intent signals



Vision Agent

Object detection and environmental context via **YOLO**



Coordinator Agent

Multimodal fusion to resolve ambiguities and infer true intent



Memory Agent

Contextual history for personalized, consistent behavior



Safety Agent

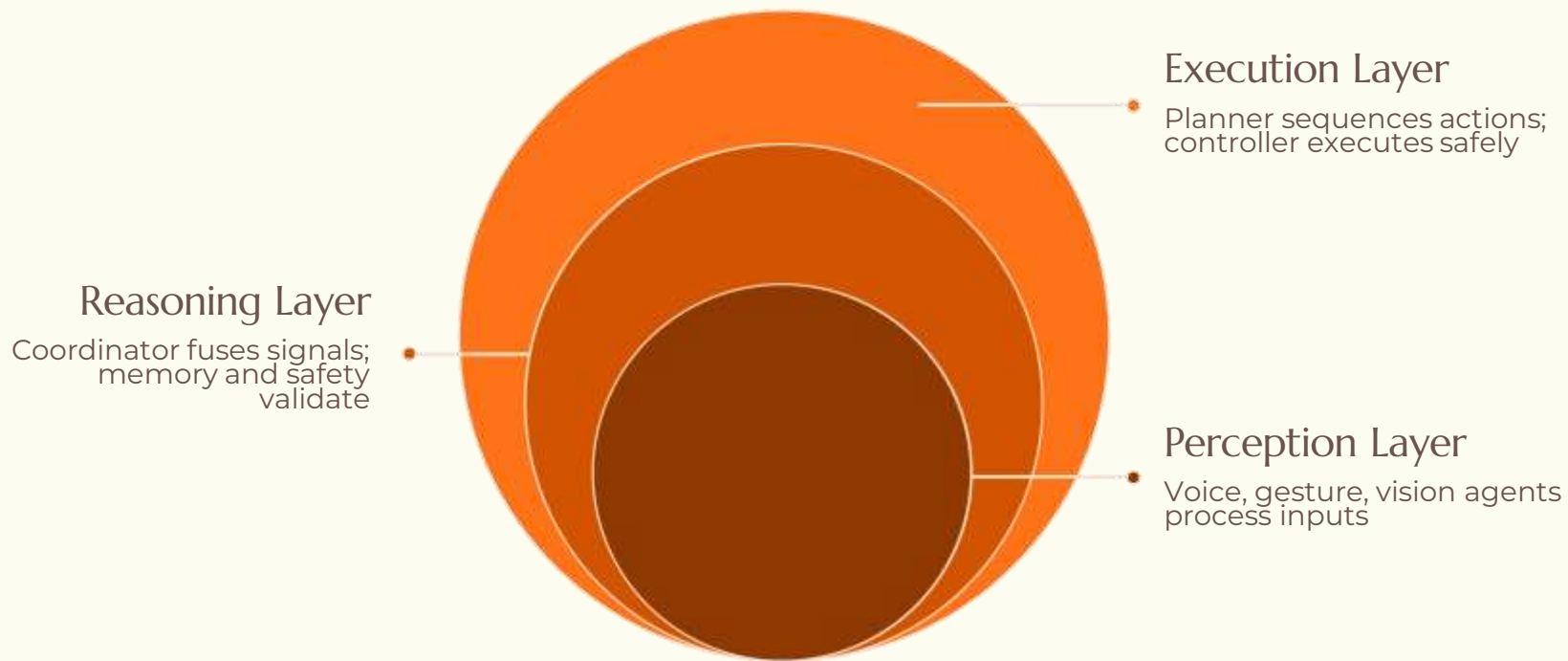
Pre-execution verification to prevent unsafe actions



Task Planner

Converts validated intent into executable robot actions

System Architecture: Three Integrated Layers



Each layer operates in sequence—raw sensory inputs are processed in parallel at the perception layer, unified and validated by the reasoning layer, and finally translated into safe, executable robot actions by the execution layer.

Key Innovation: Multimodal Fusion + Safety-Aware Reasoning

Fusion Resolves Ambiguity

Single modality fails under real-world noise. Multimodal fusion resolves **70%+ of ambiguous cases** by cross-validating voice, gesture, and vision signals.

Safety-Aware Execution

The Safety Agent evaluates environmental conditions **before any action executes**—preventing accidents in human-robot proximity zones.

LLM-Powered Reasoning

The Coordinator Agent applies large language model reasoning to weight signals and **validate intent** before committing to any task plan.

Why This Matters: Closing the Research Gap

→ Voice-Only Systems

Lack gesture and visual context integration—fail in occluded or noisy scenarios

→ Existing Multi-Agent Frameworks

Lack multimodal human communication channels and real-time safety validation layers

→ Gesture Recognition Systems

Perform perception only—no high-level semantic reasoning or intent validation

→ Our Contribution

First unified system combining all seven capabilities—multimodal perception, contextual memory, task planning, and safety-aware reasoning—in real-time HRI

Technical Implementation: Feasible Within Constraints

Models & Stack

Perception Models

Whisper-small · MediaPipe · **YOLOv8** — lightweight for real-time inference

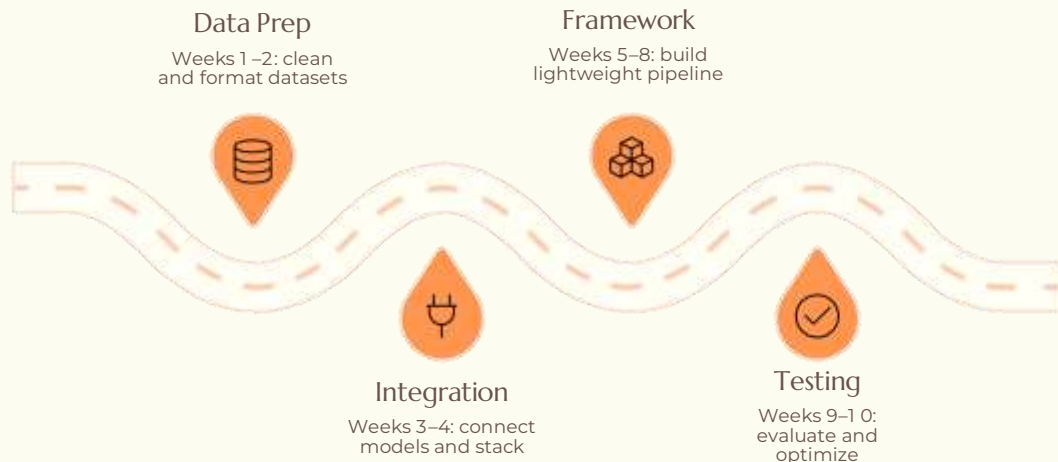
Software Stack

Python · PyTorch · **ROS2** · OpenCV

Hardware

Intel i7 + **RTX 4060 GPU** — optimized for on-device inference

Development Timeline



Validation & Next Steps



Datasets

Mozilla Common Voice · HaGRID · COCO · custom multimodal samples



Baselines

Voice-only · gesture-only · vision-only · multimodal without safety layer



Metrics

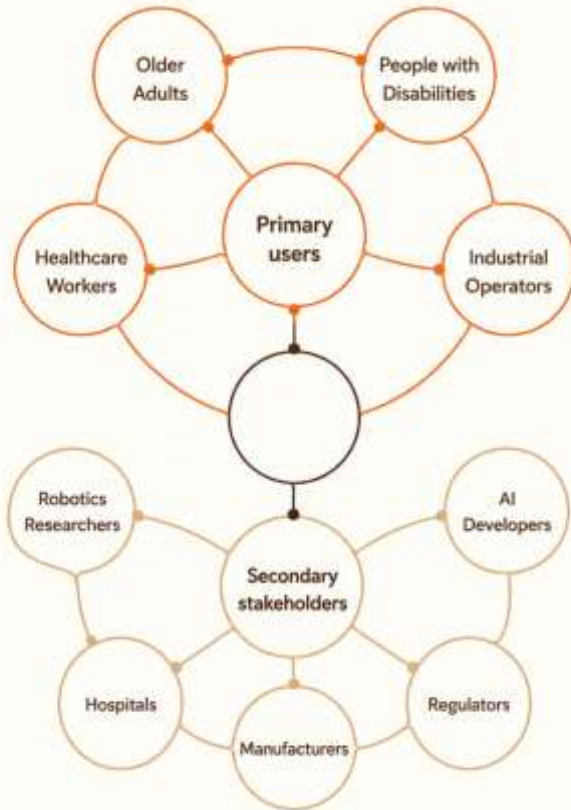
Intent accuracy · task completion rate · latency · safety violation rate



Deployment

User studies with healthcare & manufacturing stakeholders in real environments

Stakeholders & Impact



Primary Beneficiaries

Older adults and people with disabilities gain intuitive assistive interaction. Healthcare workers and industrial operators benefit from safer, more reliable collaborative robots.

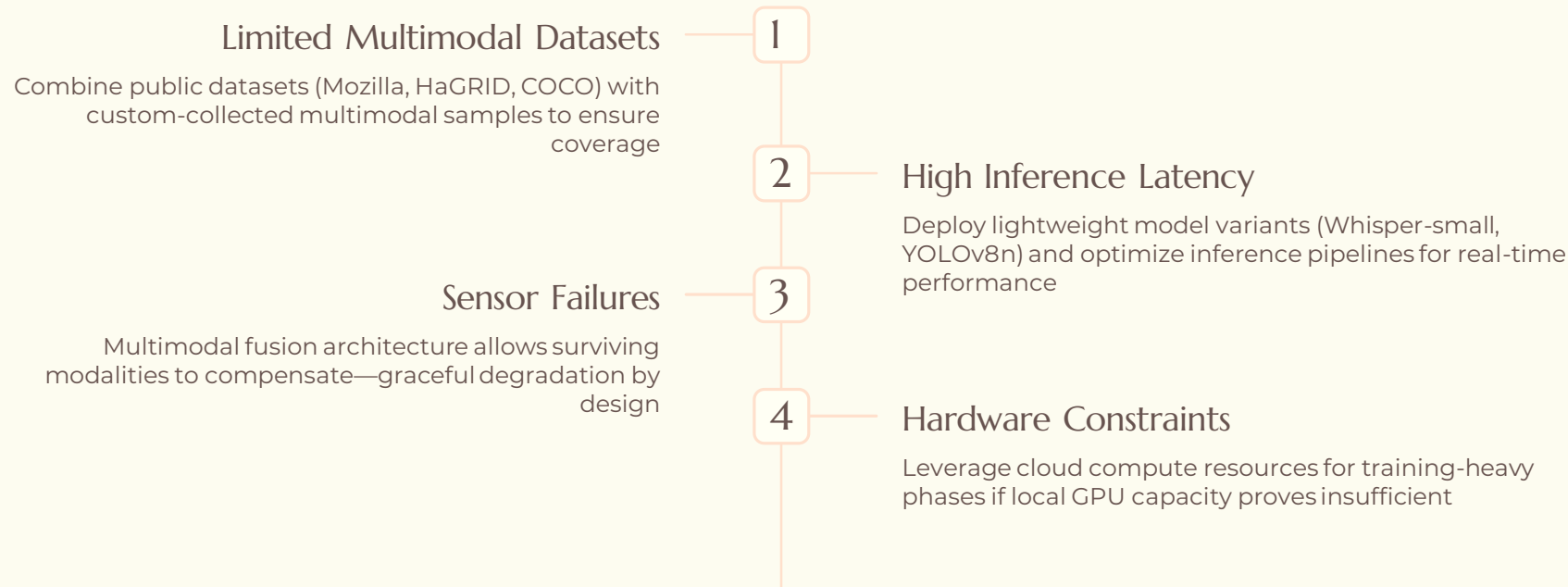
Secondary Stakeholders

Robotics researchers, AI developers, hospitals, manufacturers, and regulators all benefit from a validated, open framework for safe multimodal HRI.

Cross-Sector Use Cases

- Rehabilitation and assisted living
- Precision manufacturing and logistics
- Home assistance and elder care

Risks & Mitigation



 Resilience is a design principle, not an afterthought — every risk has a structural mitigation built into the framework architecture.